

ASSIGNMENT #01

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, January 23rd TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

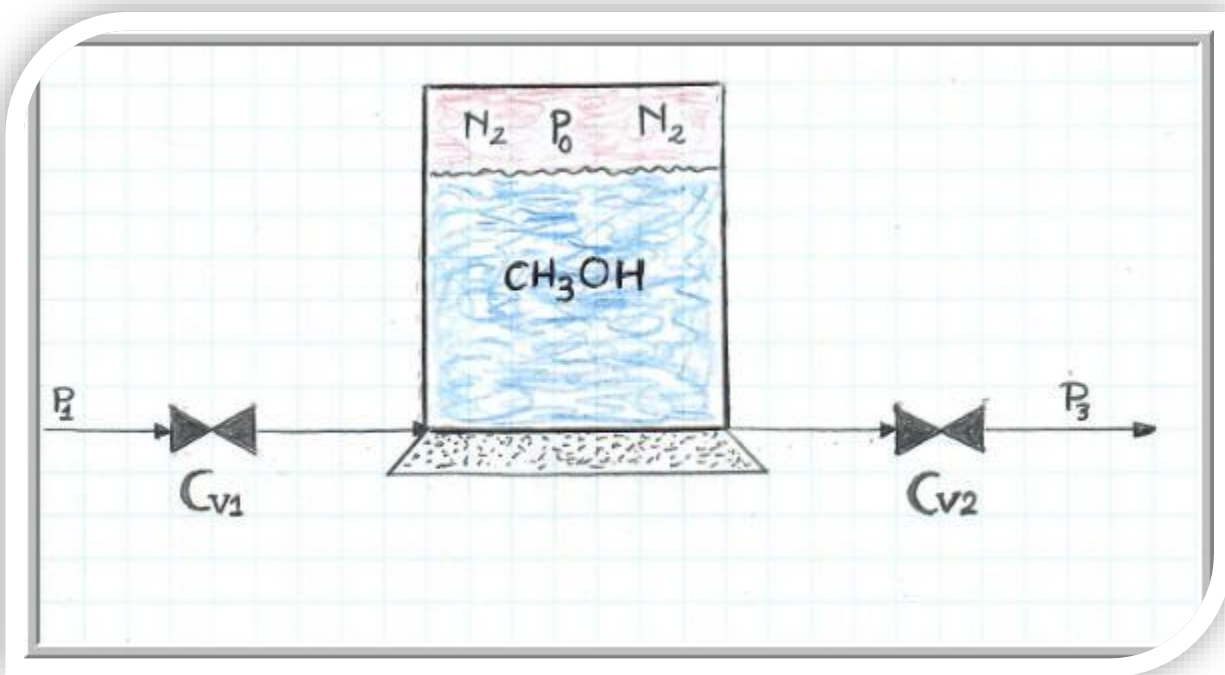
Problem A: Modeling of a Blanketed Storage Tank for Methanol

A **well-insulated** cylindrical tank (see Figure below) is used for storage of liquid methanol. The tank has volume of $V = 5000 \text{ ft}^3$ and the surface area of the bottom is $A = 200 \text{ ft}^2$. The liquid methanol is blanketed with nitrogen to avoid contamination with air and moisture. The feed and effluent line valve coefficients are $C_{v1} = C_{v2} = 110 \text{ gpm} / \sqrt{\text{psi}}$. The valves exhibit linear characteristics and the flow rate through the valves is given by the equations

$$F_1 = C_{v1} \sqrt{\frac{\Delta P_1}{s_g}} \quad \text{and} \quad F_2 = C_{v2} \sqrt{\frac{\Delta P_2}{s_g}}$$

where s_g is the specific gravity of methanol. The gauge pressure in the tank (measured in the nitrogen space) is $P_0 = 5 \text{ psig}$ and the temperature $T_0 = 25^\circ \text{C}$. The upstream to the feed line valve pressure is $P_1 = 33.5 \text{ psia}$ and the downstream to the effluent line valve pressure is $P_3 = 20.5 \text{ psia}$.

Figure: Methanol Storage Tank



Develop the necessary equation(s) required to monitor the liquid methanol height and the temperature of the vapor space as a function of time, due to various pressure disturbances in the feed and effluent lines. After you develop the model perform the following tasks

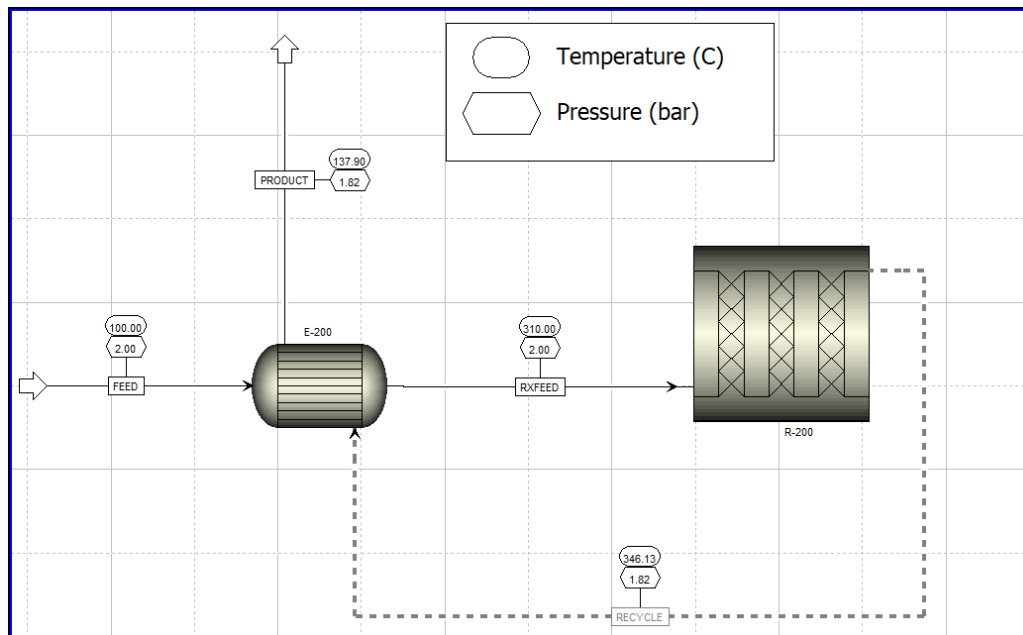
1. Calculate the steady-state liquid methanol height in **ft**
2. Calculate the number of moles of methanol and nitrogen in the vapor space
3. Perform a dynamic simulation for about 4 hr to demonstrate that the liquid height and temperature remain constant at the steady-state values in the absence of any disturbances
4. Start the simulation again at $t=0$ min. At time $t=2$ hr, the upstream to the feed valve pressure increases by 20%. Plot the liquid height and vapor space temperature as a function of time and calculate the new steady-state values.
5. Calculate the final pressure in **psia** of the vapor space in the storage tank
6. Calculate the time, in **min**, required to reach the new steady-state values defined as 99.5% of the asymptotic (steady-state) values.
7. In one Appendix of your report include all the plots of your analysis.
8. In a second Appendix of your report include the computer code with all the data that you used to do the numerical integration. You are free to use any software (like Matlab) but I would strongly recommend that you use the CHE Calculator

Data and Assumptions:

1. You may get the methanol and nitrogen properties from the ASPEN Plus simulator or the CHE Calculator (I would recommend the CHE Calculator)
2. Assume that nitrogen/methanol mixture in the vapor space behaves as an ideal gas with constant heat capacities C_p and C_v .
3. Assume that the amount of methanol in the vapor space does not change appreciably because of temperature and pressure variations.

Problem B: Design vs. Operational Models: Understanding Positive Feedback

Consider the ASPEN Plus simulation file that is uploaded on the UIC Blackboard along with your first assignment. The simulation file describes the catalytic oxidation of a hydrocarbon in an adiabatic fixed bed reactor. The feed to the reactor is preheated to a temperature of 310 °C using a CFE (Combined Feed Exchanger) as shown in the diagram below:



Perform the following tasks:

1. Study the simulation file and provide a description (no longer than one page) of what this design is all about with some detail on the reaction mechanism and the reactor design.
2. Modify the simulation file to reflect operational mode. The current one is the design mode.
3. Determine what will happen if due to some unforeseen disturbances the reactor effluent temperature increases by 2 °C. When answering this question assume that there are no controls in place. Describe in detail your answer and corroborate it with the proper simulations. Assuming that it takes 20 seconds for the disturbance to go through the process, plot the temperature of the reactor effluent versus time for the first 60 seconds after the onset of the disturbance.
4. Recommend a solution to remedy the situation resulting from the unexpected rise in temperature and modify the simulation file to reflect your recommendation. Describe in detail your recommendation and report the results of your simulation that reflect your recommendation. Also calculate the net increase or decrease in utilities after the onset of the disturbance.

Chemical Process Control Report 1

Kyle Hankosky

January 2026

1 Problem A: Modeling of a Blanketed Storage Tank for Methanol

The dynamic balances are developed using a standard accumulation framework. For the liquid phase, the general balance reduces to a volume balance since no generation or consumption occurs:

$$\text{Accumulation} = \text{In} - \text{Out}$$

The inlet and outlet volumetric flow rates are modeled using valve equations,

$$F_1 = C_{v1} \sqrt{\frac{\Delta P_1}{s_g}}, \quad F_2 = C_{v2} \sqrt{\frac{\Delta P_2}{s_g}}$$

The rate of change of liquid volume is therefore

$$\frac{dV}{dt} = F_1 - F_2$$

For a tank with constant cross-sectional area A , the liquid volume is $V = AH$, giving

$$A \frac{dH}{dt} = F_1 - F_2 \quad \Rightarrow \quad \frac{dH}{dt} = \frac{F_1 - F_2}{A}$$

The vapor-phase energy balance is written using the first law for a closed system,

$$\frac{dU}{dt} = \dot{Q} - \dot{W}$$

Under adiabatic conditions ($\dot{Q} = 0$), and assuming only boundary work,

$$\frac{dU}{dt} = -P_0 \frac{dV_g}{dt}$$

For nitrogen modeled as an ideal gas,

$$U = n_{N_2} C_v T \quad \Rightarrow \quad \frac{dU}{dt} = n_{N_2} C_v \frac{dT}{dt}$$

Combining expressions yields

$$\frac{dT}{dt} = -\frac{P_0}{n_{N_2} C_v} \frac{dV_g}{dt}$$

The gas volume is related to the liquid height by

$$V_g = V_{\text{tank}} - AH \quad \Rightarrow \quad \frac{dV_g}{dt} = -A \frac{dH}{dt}$$

Substituting the liquid balance gives the final coupled temperature equation,

$$\frac{dT}{dt} = \frac{P_0}{n_{N_2} C_v} (F_1 - F_2)$$

The governing differential equations are therefore

$$\frac{dH}{dt} = \frac{F_1 - F_2}{A}, \quad \frac{dT}{dt} = \frac{P_0}{n_{N_2} C_v} (F_1 - F_2)$$

1.1 Steady-State Methanol Liquid Height Calculation

At steady state, the liquid height is constant in time, so

$$\frac{dH}{dt} = 0$$

From the mass balance,

$$\frac{dH}{dt} = \frac{F_1 - F_2}{A} = 0 \Rightarrow F_1 = F_2$$

Using the valve flow relations and including the hydrostatic contribution to the liquid-side pressure,

$$F_1 = C_{v1} \sqrt{\frac{P_1 - (P_0 + \rho GH)}{s_g}}, \quad F_2 = C_{v2} \sqrt{\frac{(P_0 + \rho GH) - P_2}{s_g}}$$

Thus, the steady-state height is obtained by solving

$$C_{v1} \sqrt{\frac{P_1 - (P_0 + \rho GH)}{s_g}} = C_{v2} \sqrt{\frac{(P_0 + \rho GH) - P_2}{s_g}}$$

Using Goal Seek and the following parameters, the steady-state liquid height was determined to be

$$H_{ss} = 21.24 ft$$

Variable [unit]	Value
P_0 [psia]	19.7
P_1 [psia]	33.5
P_2 [psia]	20.5
s_g [-]	0.793
A [ft ²]	200
V [ft ³]	5000
C_p [Btu/(lbmol·°R)]	3863
ρ [lb/ft ³]	49.5
G [psi/ft]	0.006944
T_{init} [°R]	536.67

Table 1: Model parameters used in steady-state and dynamic calculations

1.2 Moles of Methanol and Nitrogen in the Vapor Space

After steady state is achieved, the liquid height remains constant at H_{ss} . The vapor-space height is therefore determined from the tank geometry as

$$H_g = H_{\text{tank}} - H_{ss}$$

Using the constant cross-sectional area of the tank, the vapor-space volume is

$$V_g = AH_g$$

At steady state, the total pressure in the vapor space is known from the pressure balance. Methanol in the headspace is assumed to be in equilibrium with the liquid phase and therefore exerts its saturated vapor pressure at the system temperature,

$$P_{\text{MeOH}} = P_{\text{sat, MeOH}}(T)$$

The remaining pressure in the vapor space is attributed to nitrogen, giving

$$P_{N_2} = P_{\text{total}} - P_{\text{MeOH}}$$

Assuming ideal gas behavior for both vapor-phase species, the number of moles of each component in the vapor space is calculated using the ideal gas law. For nitrogen,

$$n_{N_2} = \frac{P_{N_2} V_g}{RT}$$

and for methanol vapor,

$$n_{\text{MeOH}} = \frac{P_{\text{MeOH}} V_g}{RT}$$

Component	P [psia]	n [mol]
Nitrogen (N_2)	17.398	2.274
Methanol (MeOH)	2.302	0.301
Total	19.7	2.575

Table 2: Vapor-space partial pressures and moles at steady state

These expressions provide the steady-state quantities of nitrogen and methanol in the vapor space, which are used as initial conditions for the subsequent dynamic simulation.

1.3 Four-Hour Dynamic Simulation Without Disturbance

The system of coupled ordinary differential equations derived in the previous section was solved to evaluate the transient behavior of the blanketed methanol storage tank under nominal operating conditions. The governing equations for liquid height and vapor-space temperature,

$$\frac{dH}{dt} = \frac{F_1 - F_2}{A}, \quad \frac{dT}{dt} = \frac{P_0}{n_{N_2} C_v} (F_1 - F_2),$$

were implemented directly using the valve flow relations that include the hydrostatic pressure contribution.

The equations were integrated over a four-hour period using the CHE Calculator Runge–Kutta 5 (RK5) method in conjunction with the custom `fODE` function. Steady-state values of liquid height and vapor-space temperature, obtained from the analytical condition $F_1 = F_2$, were used as the initial conditions. All pressures, valve coefficients, and physical properties were held constant throughout the simulation, and no external disturbances were applied.

The resulting time histories for liquid height and vapor-space temperature show that both state variables remain constant over the four-hour simulation window, aside from negligible numerical transients at the start of integration. This confirms that the steady-state solution satisfies the governing equations and that the numerical implementation is stable. The corresponding plots are provided in Appendix A.1, and the VBA implementation of the governing equations is included in Appendix B.1.

The steady-state values obtained from the dynamic simulation are summarized below.

Quantity [unit]	Steady-state value
Liquid height, H_{ss} [ft]	21.23636
Vapor-space temperature, T_{ss} [°R]	539.493

Table 3: Steady-state results obtained from the four-hour dynamic simulation without disturbance

1.4 Four-Hour Dynamic Simulation With Disturbance

The dynamic response of the system to an imposed disturbance was evaluated by modifying the governing equations to account for a step change in operating conditions. The same coupled mass and energy balance equations used in the nominal case were retained; however, the inlet pressure and vapor-space pressure were allowed to vary in time to represent the disturbance.

A step increase of 20% in the methanol feed pressure was introduced at $t = 120$ minutes. For $t \geq 120$ minutes, the inlet pressure P_1 was increased, directly affecting the inlet flow rate. Following the disturbance, the nitrogen blanket pressure was no longer treated as constant, but was instead recalculated at each time step using the ideal gas relation based on the fixed nitrogen in the head space. Prior to the disturbance, the nitrogen was calculated from the initial steady-state conditions.

The modified equations were integrated over a four-hour period using the CHE Calculator Runge–Kutta 5 (RK5) solver and the disturbance-specific `fODE_D` function. This formulation allows the inlet flow rate to respond immediately to the pressure increase, while the outlet flow rate adjusts through the hydrostatic pressure dependence on liquid height.

The resulting time histories show a transient increase in liquid height following the disturbance, accompanied by corresponding changes in vapor-space temperature and pressure. As the system evolves, the inlet and outlet flow rates re-equilibrate at a new operating point, and the state variables asymptotically approach a new steady state. The corresponding plots are provided in Appendix A.2, and the VBA implementation of the disturbance logic is included in Appendix B.2.

Quantity [unit]	Steady-state value Post Disturbance
Liquid height, H_{ss} [ft]	21.76183
Vapor-space temperature, T_{ss} [°R]	538.847

Table 4: Steady-state results obtained from the four-hour dynamic simulation with disturbance

1.5 Vapor Space Pressure Following the Disturbance

Following the disturbance, the vapor-space pressure was calculated using the final steady-state liquid height and vapor-space temperature obtained from the dynamic simulation. The vapor-space height is determined from the tank geometry as

$$H_g = H_{\text{tank}} - H_{ss} = 25 - 21.76183 = 3.23817 \text{ ft}$$

Using the constant cross-sectional area, the vapor-space volume is

$$V_g = AH_g = 200(3.23817) = 647.63 \text{ ft}^3$$

The nitrogen partial pressure in the vapor space was calculated using the ideal gas law,

$$P_{N_2} = \frac{n_{N_2}RT}{V_g}$$

with $n_{N_2} = 2.27393 \text{ lbmol}$, $T = 538.847 \text{ °R}$, and $R = 10.7316 \text{ psia}\cdot\text{ft}^3/(\text{lbmol}\cdot\text{°R})$, giving

$$P_{N_2} = 20.30 \text{ psia}$$

Methanol was assumed to be in equilibrium with the liquid phase, and its vapor pressure at the final temperature was taken as

$$P_{\text{MeOH}} = 2.50 \text{ psia}$$

The total vapor-space pressure following the disturbance is therefore

$$P_{\text{total}} = P_{N_2} + P_{\text{MeOH}} = 20.30 + 2.50 = 22.80 \text{ psia}$$

1.6 Time Required to Reach the New Steady State

The time to reach the new steady state was defined as the first time t that the remaining deviation from the final steady-state value satisfies

$$\frac{|x(t) - x_{ss}|}{|x_{ss} - x_0|} \leq 0.005$$

where x_0 is the value immediately after the disturbance and x_{ss} is the final asymptotic value. Using the simulation results, this criteria was met at $t \approx 4 \text{ min}$, which is reported as the settling time to 99.5% of the new steady state.

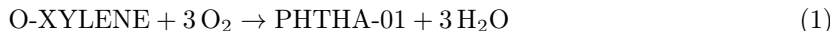
2 Problem B: Design vs. Operational Models: Understanding Positive Feedback

2.1 Process Description and Reactor Design

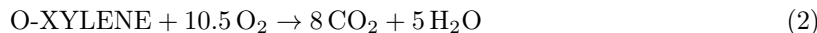
This Aspen Plus simulation represents a gas-phase catalytic oxidation process for producing phthalic anhydride from a dilute aromatic feed. The primary reactant is *o*-xylene, introduced at low concentration with air. The process flowsheet consists of a feed stream preheated in a feed-effluent heat exchanger, followed by an adiabatic plug flow reactor. The hot reactor effluent passes back through the exchanger to recover sensible heat before exiting the system. The simplified layout emphasizes reactor performance and heat integration rather than downstream separations.

The reactor is modeled using an RPlug block configured as a multitubular packed-bed reactor operating in the vapor phase only. The reactor contains 99,999 tubes, each 0.5 m in length and 0.025 m in diameter. A total of 30,000 kg of catalyst is distributed throughout the bed, with a specified void fraction of 0.38 and a catalyst particle diameter of 0.003 m. The reactor operates adiabatically, with no external heat removal, making reaction heat release and feed dilution the primary means of temperature control.

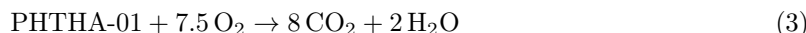
The reaction network is implemented using lumped power-law kinetics to represent the dominant oxidation pathways. The primary desired reaction is the selective oxidation of *o*-xylene to phthalic anhydride:



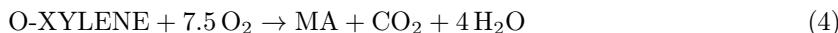
Several side reactions are included to account for parallel and consecutive over-oxidation. These include direct combustion of *o*-xylene,



consecutive oxidation of phthalic anhydride,



and a parallel pathway producing maleic anhydride:



All reactions are modeled as irreversible vapor-phase reactions with partial-pressure driving forces and rates expressed per unit catalyst weight. For the main phthalic anhydride formation reaction, first-order dependence on both *o*-xylene and oxygen is assumed, with Arrhenius parameters of $k = 0.0206$, $E = 27$ kcal/mol, and a reference temperature of 300 °C.

The reactor feed consists primarily of nitrogen and oxygen, with an *o*-xylene mole fraction of approximately 0.007 and trace amounts of other aromatics. This high level of inert and oxidant dilution limits the adiabatic temperature rise and reflects typical industrial practice for aromatic oxidation reactors. At the reactor outlet, the model predicts production of approximately 1,274 kg/hr of phthalic anhydride, along with 426.6 kg/hr of carbon dioxide, 43.6 kg/hr of maleic anhydride, and 585.9 kg/hr of water, with excess oxygen and nitrogen remaining in the effluent.

2.2 Modification of the Simulation to Operation Mode

The model was checked to ensure it was fully specified in Aspen by verifying that the simulation ran without errors and that no design specifications were active. The reactor size and catalyst amount were fixed, while operating conditions were controlled through the feed stream, representing operation mode.

2.3 Effect of a Reactor Effluent Temperature Disturbance

This scenario considers an uncontrolled disturbance that results in a 2.0 °C increase in the reactor effluent temperature relative to the baseline operating condition. No control systems are assumed to be in place; therefore, the disturbance is not corrected and the reactor transitions from its original steady state to a new steady state at the elevated temperature.

The reactor kinetics are modeled using Arrhenius-type power-law expressions, meaning that reaction rate constants increase with temperature. As a result, an increase in reactor temperature leads to higher overall reaction rates. Because the reaction network includes both the desired oxidation of *o*-xylene to phthalic anhydride and multiple parallel and consecutive oxidation side reactions, the temperature increase affects all reaction pathways simultaneously. In particular, higher temperatures generally promote over-oxidation reactions, including direct combustion of *o*-xylene and oxidation of phthalic anhydride to carbon dioxide and water. Consequently, while *o*-xylene conversion may increase, selectivity to phthalic anhydride may decrease due to increased formation of CO₂ and H₂O.

To corroborate the expected behavior, two steady-state Aspen Plus simulations were performed. The first represents the baseline operating case at the nominal reactor effluent temperature. The second represents the disturbed case, where the reactor effluent temperature is increased by 2.0 °C. Because Aspen Plus is a steady-state simulator, the disturbance is evaluated by comparing the converged results of these two cases. Differences in outlet temperature and product flow rates between the baseline and disturbed simulations quantify the impact of the temperature upset.

The problem statement specifies that the disturbance requires 20 s to propagate through the process. This delay is treated as a transport dead time between the onset of the disturbance and its appearance in the reactor effluent temperature. With no control action present, the temperature response is therefore modeled as a delayed step change. For times $0 \leq t < 20$ s, the reactor effluent temperature remains at the baseline value, $T_{E,0}$. At $t = 20$ s, the temperature increases to $T_{E,0} + 2$ °C and remains at this value for the remainder of the observed period.

The resulting reactor effluent temperature profile for the first 60 s following the onset of the disturbance is shown in Appendix B.3. The plot demonstrates the specified 20 s transport delay followed by a sustained 2 °C increase in effluent temperature, consistent with an uncontrolled thermal disturbance.

2.4 Recommended Mitigation for Temperature Disturbance

The unexpected increase in reactor effluent temperature represents a thermal upset that requires corrective action. Although adding a heater downstream of the feed-effluent heat exchanger could compensate for the observed temperature decrease of approximately 10 °C, introducing additional heat upstream of an adiabatic oxidation reactor increases the risk of runaway behavior due to the highly exothermic reaction system.

A safer mitigation strategy is to install a cooler on the reactor effluent stream. This approach removes excess heat from the process without increasing the reactor inlet temperature. The Aspen Plus simulation was modified to include an effluent cooler, with the cooling duty adjusted to return the effluent temperature to its baseline value following the disturbance. The results indicate that reactor operation stabilizes and product flow rates return close to baseline conditions.

The addition of the effluent cooler results in an increase in cooling utility demand, while no additional heating utilities are required. This change represents the net utility impact associated with mitigating the temperature disturbance.

A Plots used in Q1 Analysis

A.1 Q3: Without Disturbance

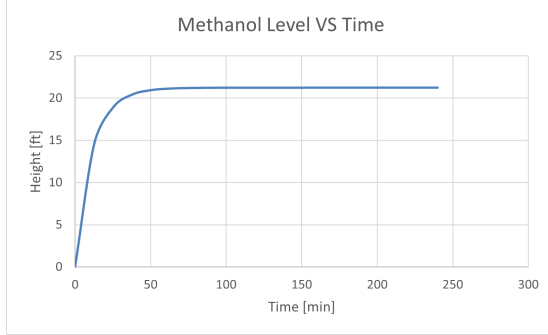


Figure 1: Liquid height vs. time

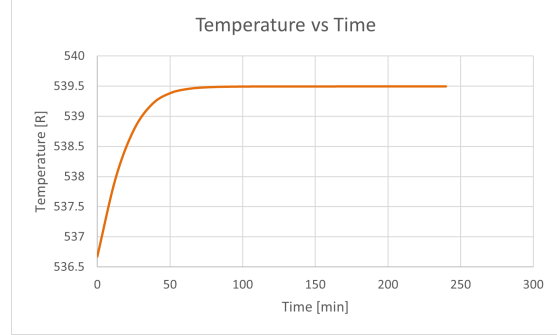


Figure 2: Vapor-space temperature vs. time

Figure 3: Four-hour dynamic simulation without disturbance

A.2 Q4: With Disturbance

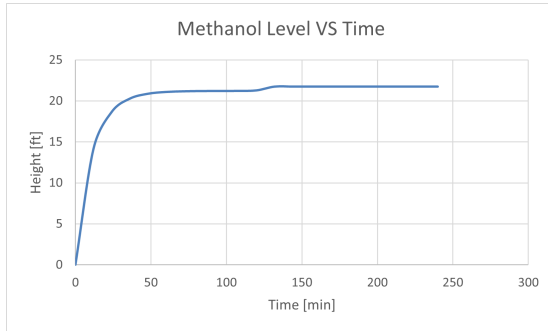


Figure 4: Liquid height vs. time

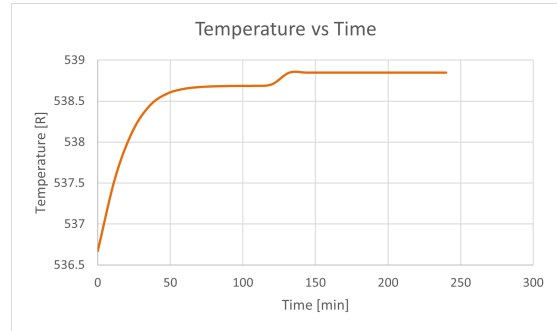


Figure 5: Vapor-space temperature vs. time

Figure 6: Four-hour dynamic simulation without disturbance

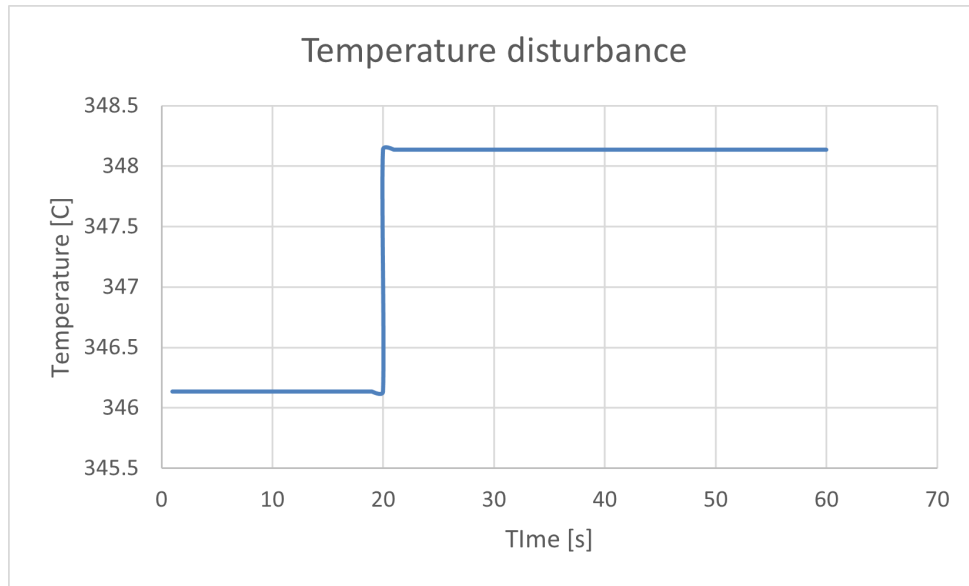


Figure 7: Reactor effluent temperature response to a +2 °C disturbance with a 20 s transport delay and no control action.

1

B VBA Code used in solution

B.1 No disturbance

```
Function fODE(T As Double, u() As Double, Optional Param As Range)
'User function for ODE Sample Problem
'*****
' du(1)/dt = 1 / A * (F1 - F2)
' du(2)/dt = (P2/n_N2/ Cv)*(F1 - F2)
'-----
Dim n As Integer
n = UBound(u)
Dim P0 As Double, P1 As Double, P3 As Double
Dim sg As Double, A As Double, V As Double, Cv As Double
Dim rho As Double, Tint As Double
Dim R As Double, G As Double

P0 = Param.Cells(1, 1) 'n2 blanket pressure
P1 = Param.Cells(2, 1) 'meoh feed pressure
P3 = Param.Cells(3, 1) 'meoh outlet pressure
sg = Param.Cells(4, 1) 'specific gravity of meoh
A = Param.Cells(5, 1) 'tank area
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```

V = Param.Cells(6, 1) 'tank volume
Cv = Param.Cells(7, 1) 'specific heat at constant volume of n2
rho = Param.Cells(8, 1) 'density of meoh
G = Param.Cells(9, 1) 'gravity
Tint = Param.Cells(10, 1) 'initial temperature
R = 10.7316 'psia*ft3/(lbmol*R)

Dim n_N2 As Double
n_N2 = P0 * (5000 - u(1) * A) / (R * Tint)

Dim F1 As Double, F2 As Double
F1 = 110 * Sqr((P1 - (rho * G * u(1) + P0)) / sg)

If (rho * G * u(1) + P0) - P3 < 0 Then
    F2 = 0
Else
    F2 = 110 * Sqr(((rho * G * u(1) + P0) - P3) / sg)
End If

Dim r1 As Double
r1 = F1 - F2

ReDim f(1 To n) As Double
f(1) = r1 / A
f(2) = (P0 / (n_N2 * Cv)) * r1

fODE = f
End Function

```

B.2 With Disturbance

```

Function fODE_D(T As Double, u() As Double, Optional Param As Range)
'User function for ODE Sample Problem
'*****
' du(1)/dt = 1 / A * (F1 - F2)
' du(2)/dt = (P2/n_N2/ Cv)*(F1 - F2)
'-----
Dim n As Integer
n = UBound(u)
Dim P0 As Double, P1 As Double, P3 As Double
Dim sg As Double, A As Double, V As Double, Cv As Double
Dim rho As Double, Tint As Double, n_N2 As Double
Dim R As Double, G As Double

P0 = Param.Cells(1, 1) 'n2 blanket pressure
P1 = Param.Cells(2, 1) 'meoh feed pressure

```

```

P3 = Param.Cells(3, 1) 'meoh outlet pressure
sg = Param.Cells(4, 1) 'specific gravity of meoh
A = Param.Cells(5, 1) 'tank area
V = Param.Cells(6, 1) 'tank volume
Cv = Param.Cells(7, 1) 'specific heat at constant volume of n2
rho = Param.Cells(8, 1) 'density of meoh
G = Param.Cells(9, 1) 'gravity
Tint = Param.Cells(10, 1) 'initial temperature
n_N2 = Param.Cells(11, 1) 'moles of n2 in blanket
R = 10.7316 'psia*ft3/(lbmol*R)

if T >= 120 Then
    P1 = P1 * 1.20
    P0 = (n_N2 * R * u(2)) / (5000 - u(1) * A)
else if T < 120 Then
    n_N2 = P0 * (5000 - u(1) * A) / (R * Tint)
end if

Dim F1 As Double, F2 As Double
F1 = 110 * Sqr((P1 - (rho * G * u(1) + P0)) / sg)

If (rho * G * u(1) + P0) - P3 < 0 Then
    F2 = 0
Else
    F2 = 110 * Sqr(((rho * G * u(1) + P0) - P3) / sg)
End If

Dim r1 As Double
r1 = F1 - F2

ReDim f(1 To n) As Double
f(1) = r1 / A
f(2) = (P0 / (n_N2 * Cv)) * r1

fODE_D = f

End Function

```